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Dadnet: dual-attention detection network for crack segmentation on tomb murals

Meng Wu^{1,2*}, Ruochang Chai^{1†}, Yongqin Zhang^{3,5†} and Zhiyong Lu^{4†}

Abstract

Many tomb murals have punctate losses, cracks, and craquelure due to underground subsidence and changes in their physical support. Visual non-destructive detection techniques enable rapid assessment of how much tomb murals are affected by cracking, providing suggestions for their conservation. However, tomb murals are typically created by sketching outlines and then colored. Detailed sketches can easily interfere with crack detection work, requiring the use of deep learning network to better learn crack features and improve detection accuracy. At the same time the limited data of tomb mural presents a challenge to build a deep learning network. To address these issues, this paper introduces a novel dual-attention detection network (DADNet) for crack segmentation of tomb murals. In this work, a customized dataset is first constructed by collecting mural images from the Tang Dynasty tombs. Then the ConvNeXt framework serves as the basis for feature extraction, enhancing the process. Lastly, a dual-attention module utilizing neighborhood attention and biaxial attention is employed to accurately identify the crack regions. Neighborhood attention performs a local self-attention operation around the pixel point, addressing the limitations of self-attention. This approach significantly reduces computational demands as the image size increases. Biaxial attention performs attention calculations in the horizontal and vertical directions. This compensates for the limitation of neighborhood attention in capturing global dependencies. Our DADNet outperformed the competing methods, achieving the highest recorded scores of 78.95% for MIoU and 61.05% for the Jaccard index.

Keywords Tomb mural, Crack segmentation, Attention module, Cultural protection

Introduction

Murals are the treasures of human cultural heritage. They inherit the cultural, religious, and social narratives of various epochs, serving as a vital foundation for scholars researching ancient history. Murals are primarily categorized into three types based on their locations: tomb murals, temple murals, and grotto murals. Murals often suffer from paint loss, powdering, damage by microorganisms, cracks, craquelure, graffiti, and scratch marks due to geological activities and human damage [1]. These diseases not only have an impact on the display of the content of the mural, but also pose a serious threat to its preservation. Consequently, the conservation and restoration of ancient murals constitute a primary focus of contemporary mural research.

Previously, mural restoration work relied mainly on manual restoration by professionals. The shortage of

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specialized personnel often resulted in slow progress in restoration efforts. At the same time, the analysis of mural damage and subsequent repair work involves manually labeling the murals [2]. Manual labeling not only proved to be time-consuming and labor-intensive but also carried a risk of causing secondary damage to the murals. The use of digital image processing technology in disease analysis will reduce damage to the mural while providing a scientific basis for future protection and restoration efforts.

Tomb murals contain intricate contextual details. For instance, the polo mural in Prince Zhang Huai's Tang Dynasty tomb vividly portrays people, horses, rocks, and trees. As shown in Fig. 1, there are numerous cracks in the mural. Such damage not only hampers conservation and restoration efforts but also diminishes the mural's aesthetic appreciation. Tomb murals exhibit less vibrant colors compared to those found in temples, with fine cracks and craquelures often merging into the mural content, thereby complicating the task of distinguishing them. How to accurately identify crack endings and craquelures in tomb murals has become a challenging subject.

Conventional image processing approaches for crack segmentation primarily utilize techniques such as morphological operations, clustering, and region growing. Spagnolo et al. [3] identified cracks by analyzing the

threshold output from morphological top-hat transformations. Cornelis et al. [4] applied semi-supervised clustering in post-processing to distinguish cracks of varying sizes and brightness levels, thereby eliminating incorrectly identified targets. Jaidilert et al. [5] utilized a seed-based region-growing algorithm and morphological manipulation to generate a crack segmentation map. Yang et al. [6] utilized multi-scale morphological techniques for edge gradient detection to outline crack edges for segmentation. Zhang et al. [7] implemented locally optimal hierarchical clustering, using a weighted average as the similarity metric, to isolate regions affected by mural diseases. Deng et al. [8] adopted multi-dimensional gradient detection, guided filtering, and tensor voting to identify mural diseases, iteratively refining the outcomes.

As deep learning advances, an increasing number of neural networks are being utilized for image segmentation tasks. By applying convolutional neural networks to the problem of mural crack segmentation, more detailed information about the cracks can be extracted, leading to a more precise segmentation. In 2015, Long et al. [9] proposed Fully Convolutional Networks (FCN), marking the first application of end-to-end convolutional networks to semantic segmentation tasks. In the same year, Ronneberger et al. [10] developed U-Net, which features fully symmetric encoder-decoder structures. These structures utilize contraction paths to capture contextual local features and expansion paths for accurate localization. Chen et al. [11] incorporated encoder-decoder structures and depth-separable convolutions into DeepLab to develop DeepLab V3+. The new model improves the localization of segmentation boundaries. Sizyakin et al. [12] proposed a rapid crack detection algorithm utilizing multimodal painting data. Mishra et al. [13] utilized the YOLOv5 algorithm to detect four types of cultural heritage diseases: color alteration, exposed bricks, cracks, and loss. Peng et al. [14] devised an intelligent labeling technique for mural fragments, using gradient-trainable Gabor filters and U-Net. The gradient-trained Gabor filters in the encoder are used to extract texture features of fragmented diseases. Incorporating residual convolution within skip connections reduces the semantic gap between the encoder and decoder. Yuan et al. [15] replaced the encoder part of U-Net with ResNet-50 and merged the skip connections by integrating feature information at various scales, thereby enhancing disease segmentation of colored paintings in the Palace Museum.

The essence of image segmentation lies in the comprehensive extraction and utilization of target feature information. Owing to the U-Net's encoder-decoder structure, which enables more efficient feature extraction and thus higher segmentation accuracy, it frequently serves as the foundational model in semantic



Fig. 1 Partial polo mural of prince Zhang Huai's tomb

segmentation tasks [16]. In 2014, Bahdanau et al. [17] introduced the attention mechanism. This mechanism filters feature information, enabling the model to ignore irrelevant noise and focus on key information. The relative ease of integrating the attention module into semantic segmentation models has led many researchers to incorporate attention mechanisms into their designs. Pan et al. [18] designed the Spatial Channel Hierarchical Network (SCHNet), replacing DANet's backbone with VGG19 and incorporating a novel pyramid attention module. Experimental findings demonstrate that SCHNet significantly outperforms DANet in the task of crack segmentation. Kang et al. [19] designed a semi-trainable representation network (STRNet) that utilizes squeeze-and excitation-based attention (SEA) to decrease model computation while preserving segmentation accuracy. Wang et al. [20] utilized a multiscale channel cross-fertilization module instead of the original jump connections. The proposed UTransNet, incorporating a Channel Transformer module into the U-Net segmentation model, aims to mitigate the negative effects linked to skipping connections. Cao et al. [21] proposed a Swin-Unet that utilizes a hierarchical Swin Transformer encoder with a moving window to extract contextual features. Xu et al. [22] proposed a deeper and more compact split-attention U-Net (DCSAU-Net). Based on the two frameworks of main feature protection and compact split-attention blocks, low-level and high-level semantic information are effectively utilized. It enables the model to better extract the information of interest from the complex background information. Cao et al. [23] proposed a mural segmentation model based on multi-scale feature fusion. The problem of losing small targets in murals because of the high sampling rate of disordered convolution is tackled by utilizing an attention-optimized residual space pyramid pooling module.

In this paper, we introduce a dual-attention detection network model by selecting the ConvNeXt framework as the backbone for extracting mural information. The weights of the extracted features are then adjusted using biaxial and neighborhood attention modules. This enables the model to accurately extract and utilize the crack feature information in the complex mural background. The main contributions of this paper are as follows:

1. A dataset of Tang Dynasty tomb murals, named TTMural, was established. Tomb mural crack segmentation is a challenging task. To achieve better segmentation results, we manually labeled the images of tomb chamber murals. Manual labeling, as opposed to traditional mask generation methods,

minimizes background interference and enhances the precision of the crack masks.

2. The ConvNeXt framework serves as the backbone network, enabling the model to extract more detailed mural crack information. This architecture enhances feature information for subsequent attention modules and reduces the risk of model overfitting.
3. The Dual Attention (DA) module has been added to the skip connection portion of the U-network. Neighborhood attention and biaxial attention are used to adjust the weights of features extracted by the backbone network, guiding the model's attention toward the crack region. These attention mechanisms are designed to enhance segmentation accuracy while reducing computational demand.

The structure of this paper is as follows. The "Methods" section describes the model structure of DADNet and the dual attention module. The "Experiments" section describes the dataset, experimental parameters, and objective evaluation metrics. The "Results and Discussion" section discusses the impact of module parameters, comparison experiments, ablation studies, and experimental design. Finally, the "Conclusion" section gives the conclusion.

Methods

Model structure

Figure 2 illustrates the network architecture of DADNet. The encoder, utilizing ConvNeXt as its backbone, extracts crack feature information from tomb murals. The DA module consists of the Neighborhood Attention and Biaxial Attention Block. This module helps the model better extract effective features of cracks and suppress invalid features.

The task of segmenting cracks in murals of burial chambers studied in this paper is a pixel-level semantic segmentation problem. In the binary segmentation of tomb murals, background pixels are labeled as 0, and crack pixels are labeled as 1. Similar challenges exist in tasks such as medical fundus vessel segmentation and highway crack segmentation. Compared to fundus blood vessel images, mural images are characterized by small cracks and complex background information. Compared with highway cracks, mural cracks have issues such as wide distribution, thin lines, and blending in with the surrounding surface. Therefore, detecting cracks in murals requires a sophisticated backbone network for efficient extraction of crack features to ensure accurate segmentation results. We employ ConvNeXt [24] as our backbone network, a model proposed by Facebook AI Research and UC Berkeley. ConvNeXt draws on ResNet50's design ideas about residual joins and convolutional blocks to

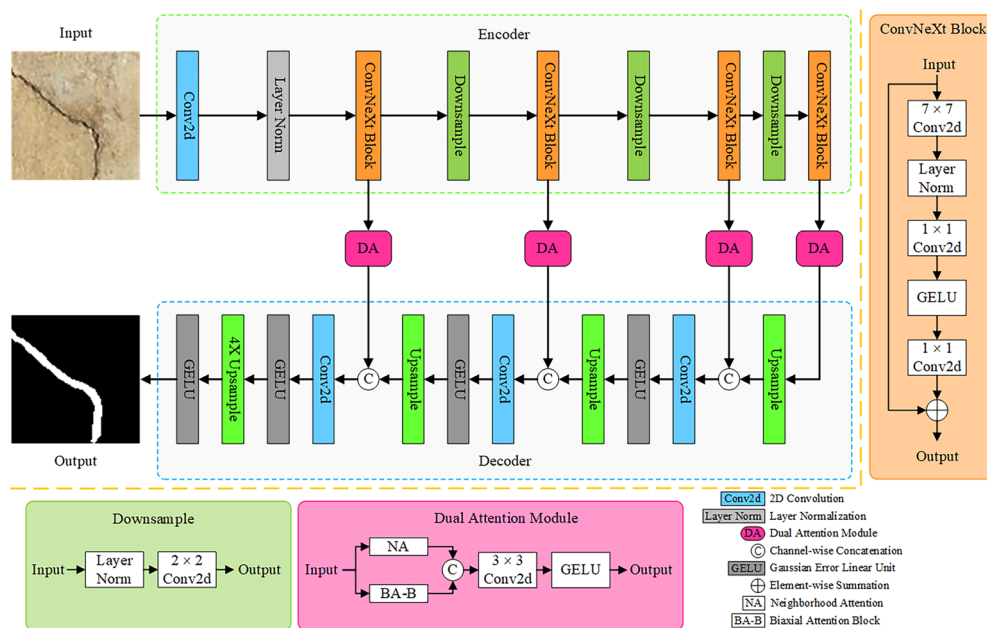


Fig. 2 Overall network structure of DADNet

build a network architecture that primarily based on convolutional operations. Meanwhile, ConvNeXt uses Layer Norm instead of Batch Norm and adjusts the convolutional kernel by referring to the design idea of Transformer. This design optimizes the performance of the network while retaining the high efficiency of the convolutional network, enabling it to achieve a TOP-1 accuracy of 87.8% on the ImageNet dataset. The ConvNeXt network is composed of ConvNeXt blocks and downsampling layers. The ConvNeXt block mainly performs deep convolutional operations. It consists of multiple 2D convolutional layers with the activation function Gaussian Error Linear Unit (GELU). Downsampling involves layer normalization and 2D convolutional layers.

The number of available images of tomb murals is limited, and the content information is complex. Consequently, it is imperative to enhance the feature extraction and utilization of both low-level structural and high-level semantic information of the cracks for the segmentation task. To prevent the backbone network's extracted features from being compromised by irrelevant features, it is considered essential to implement an attention module to weigh the feature information. By excluding invalid features, the model's performance in crack detection significantly improves.

Dual attention module

To prevent invalid features extracted by the backbone network from affecting the model, a dual attention module is incorporated into the skip connection part of the

U-Net. This module includes a neighborhood attention that focuses on the local features of the image to accurately extract crack details. The other biaxial attention block focuses on the global dependencies of the image, addressing the issue of limited sensory fields for local attention. Together, these mechanisms enhance the model's ability to effectively utilize crack features.

Neighborhood attention

To enhance the extraction of crack features, we have explored the use of the self-attention mechanism [25]. Nonetheless, the self-attention mechanism is burdened by significant computational overhead, necessitating further refinement. The significant computational demands of self-attention mainly arise from the fact that its key and value projections are the same size as the input image. This results in computational costs that escalate to the square of the image's total pixel count after matrix multiplication. Consequently, as the size of the input image increases, the computational requirements surge correspondingly. To solve this problem, it is considered beneficial to allow each pixel to conduct local self-attention computation only with some of its neighboring pixels. Therefore, Neighborhood Attention (NA) is constructed. The schematic diagram of neighborhood attention is shown in Fig. 3.

Given an input $X \in R^{H \times W \times C}$, it is a matrix with height H , width W , and channel C . Q , K and V are linear projections of X . The sizes of K and V is $k \times k \times C$, which

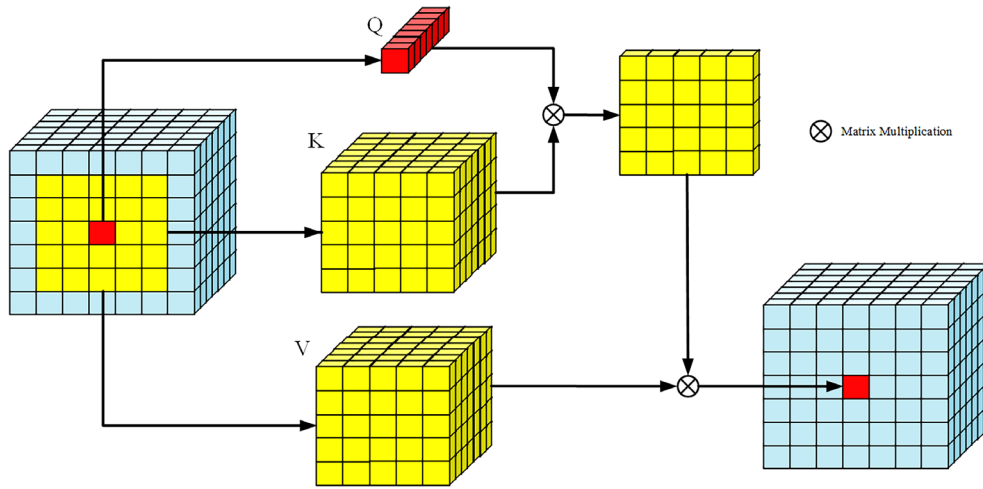


Fig. 3 Neighborhood attention

is determined by the neighborhood range k . When the neighborhood size is $k \times k$, the attentional weight $A_i^{k \times k}$ of the i -th input is the dot product of the query projection Q_i of that input and the key projection $K_{\rho_{k \times k}(i)}$ of its $k \times k$ neighbors:

$$A_i^{k \times k} = \begin{bmatrix} Q_i K_{\rho_1(i)}^T \\ Q_i K_{\rho_2(i)}^T \\ \vdots \\ Q_i K_{\rho_{k \times k}(i)}^T \end{bmatrix}. \quad (1)$$

where $\rho_{k \times k}(i)$ denotes the $k \times k$ -th nearest neighbor of the i -th input. The matrix projecting the $k \times k$ nearest neighbor projections of the i -th input is defined as the neighborhood projection matrix $V_i^{k \times k}$.

$$V_i^{k \times k} = \begin{bmatrix} V_{\rho_1(i)}^T & V_{\rho_2(i)}^T & \cdots & V_{\rho_{k \times k}(i)}^T \end{bmatrix}^T. \quad (2)$$

Then, the NA of the i -th point with neighborhood size $k \times k$ is defined as:

$$NA_{k \times k}(i) = \text{softmax} \left(\frac{A_i^{k \times k}}{\sqrt{C}} \right) V_i^{k \times k}. \quad (3)$$

where $\sqrt{C}$ is the scaling parameter and C is the dimension. This is repeated for each pixel in the feature map.

From the definition, it can be seen that as k grows, $A_i^{k \times k}$ approaches the self-attention weights and $V_i^{k \times k}$ approaches V_i itself. For an image of size $H \times W$, the amount of computation is $\frac{k^2}{H \times W}$ of self-attention.

Biaxial attention block

The special structure enables the NA to focus its attention solely on the immediate neighborhood of each pixel

point. While the NA module effectively captures low-level details within the image, it falls short in extracting high-level semantic information. To enhance the model with high-level semantic information without significantly increasing computational demands, axial attention [26] is utilized. Axial attention is proficient at capturing the global context surrounding a pixel but is less effective at discerning local pixel-level details. By integrating both kinds of attention, the model can simultaneously extract feature maps enriched with both high-level semantic and low-level detail information. Since only non-local operations are performed on a single axis for each pixel point, this attention mechanism requires only a low amount of computation.

Axial attention does not alter the original shape of the multidimensional data tensor X . It operates sequentially on a single axis of tensor X . Attention operations, denoted by AA , are executed sequentially on one axis of the tensor. AA_i represents axial attention along the i -axis of tensor X . This process integrates information along the i -axis, while preserving its independence from other axes. Axial attention on the i -axis can be achieved by aligning all axes parallel to the i -axis with the batch axis. The formula for calculating axial attention is shown below:

$$AA_i(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_i}} \right) V. \quad (4)$$

where Q , K , and V represent the query projection, key projection, and value projection respectively and d_i is the dimension of the i -axis in the key projection. Figure 4 illustrates the schematic diagram of axial attention.

On this basis, Biaxial Attention (BA) is proposed for extracting high-level semantic information from images. Attention calculations are performed separately on the

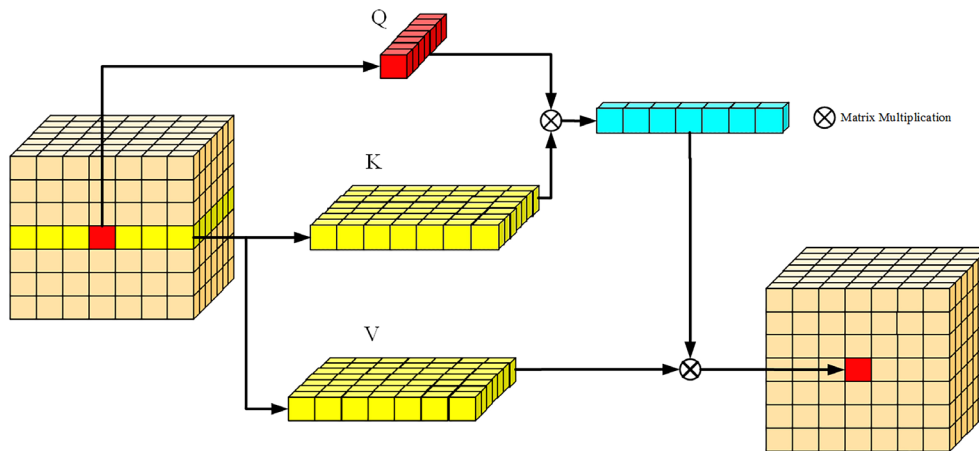


Fig. 4 Axial attention

horizontal and vertical axes, and the results are then added to the original data in the original axial order to generate a feature map. This method utilizes the feature information from both axial directions to constrain the feature map, effectively mitigating the occurrence of artifacts. The formula is shown below:

$$BA(X) = X + AA_{row}(X) + AA_{col}(X). \quad (5)$$

where X , $AA_{row}(X)$, and $AA_{col}(X)$ represent the input matrix, row attention matrix, and column attention matrix, respectively.

The BA assists the model in extracting high-level semantic information that is overlooked by the NA. Nevertheless, the encoder section of the U-Net model, characterized by its extensive number of channels, is susceptible to the loss of detailed information. We designed a biaxial attention block (BA-B) based on BA, which is similar in structure to the Receptive Field Block (RFB) [27]. The structural details of BA-B are depicted in Fig. 5. The BA-B simulates the human visual receptive field and extends the model's receptive field by performing convolution operations with

various expansion coefficients on the input data. This enables the decoder to integrate features across multiple scales from various layers effectively.

Experiments

Dataset

Painted on the west wall of the tomb of Prince Zhang Huai in the Tang Dynasty, the polo mural depicts more than twenty steeds. The knights are dressed in narrow-sleeved robes of various colors, with black boots and turbans on their heads. Grasping the reins in their left hands and wielding crescent-shaped bow staffs in their right, they present a competition scene that is both marvelous and intense. This mural illustrates the polo matches of the Tang Dynasty era and serves as a tangible testament to the cultural exchange between China and Persia.

To create the TTMural dataset, we used mural images captured with a Swiss Sinar P2 large format technical camera and a Sinar 75LV digital back. The collected mural images were divided into 21790 chunks according to the size of 224×224 pixels. After eliminating the chunks without cracks and craquelure, the computer was used to randomly select 1000 chunk images among them. The cracked

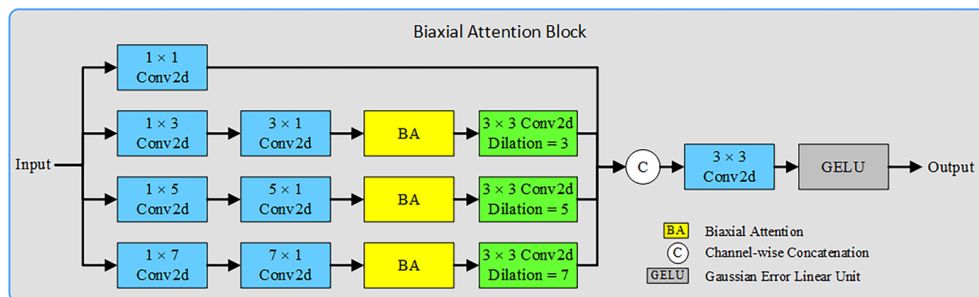


Fig. 5 Biaxial attention block

and craquelure areas were then manually labeled using LabelMe labeling software under the guidance of cultural relics experts. The respective mask images of the images were generated from the labeled files. These mural chunk images and their corresponding mask images together constitute the TTMural dataset. The images in the dataset are then randomly assigned using computerized random numbers to avoid human selection interfering with the division of the dataset. Out of this 80% is used for training set, 10% for validation set and 10% for testing set. The dataset settings are shown in Table 1:

Experimental parameters

Utilizing the PyTorch framework, the models were trained on an NVIDIA GeForce RTX 3090 graphics card. The input resolution of the data was set to 224×224 . The selected training optimizer was Adam with a β value of 0.9. The batch size was set to 16, the initial learning rate was set to $1e^{-4}$, and cross-entropy loss was employed as the loss function.

Objective evaluation metrics

Five objective evaluation metrics were used to validate the performance of the model, including Precision (P), Recall (R), F_1 -Score (F_1), Mean Intersection over Union ($MIoU$), and Jaccard distance ($Jaccard$). The equations are as follows:

$$P = \frac{TP}{TP + FP}. \quad (6)$$

$$R = \frac{TP}{TP + FN}. \quad (7)$$

$$F_1 = \frac{2PR}{P + R}. \quad (8)$$

$$MIoU = \frac{1}{k+1} \sum_{i=0}^k \frac{TP}{FN + FP + TP}. \quad (9)$$

$$Jaccard = \frac{TP}{TP + FP + FN}. \quad (10)$$

where k represents the number of categories, True Positive (TP) refers to true positive samples correctly categorized as positive, False Positive (FP) refers to false

positive samples incorrectly categorized as positive, True Negative (TN) refers to true negative samples correctly categorized as negative, and False Negative (FN) represents false negative samples incorrectly categorized as negative. P represents the percentage of all samples predicted as positive that are correct. R represents the probability of correctly predicting a positive sample. F_1 represents the harmonic mean that balances both P and R . When both variables are larger, the F_1 value increases proportionally. In semantic segmentation problems, $MIoU$ represents the average ratio of the intersection of the true labels and predicted values of each class to the union. $Jaccard$ represents the similarity between the true and predicted samples.

Experimental design

Comparative experiments

In order to evaluate the performance of DADNet in crack segmentation, five models, U-Net, DeepLab V3+, DCSAU-Net, Swin-Unet, and UCTransNet, are selected for comparison experiments with DADNet. The experiment uses the TTMural dataset and adopts the five objective evaluation metrics introduced in the objective evaluation metrics to comprehensively compare the performance of DADNet and the other five models. In order to ensure the fairness of the experiment, all models are experimented with the same dataset and configuration parameters. To avoid chance, all models are trained and tested ten times. After removing the best and worst results of $MIoU$, the average of the remaining eight results was calculated as the final score.

Ablation study

In order to deeply study the impact of ConvNeXt, NA and BA-B on the DADNet network, we performed ablation studies on the TTMural dataset using U-Net as the baseline model. Using the same experimental parameters and dataset, different combinations of each of these three modules were added to the benchmark model for the experiments. From there, we analyze the contribution of these three modules to the performance of the DADNet model.

Results and discussion

Effects of module parameters

ConvNeXt is divided into several versions with different structures based on the number of channels and modules per layer. We conducted experiments using different versions of ConvNeXt while keeping other parameters consistent. Table 2 presents the results of these experiments.

In Table 2, "channel number" refers to the count of input channels per layer across different ConvNeXt versions, and "module number" denotes the times with

Table 1 Dataset settings

Dataset	Training set	Validation set	Testing set
TTMural	800	100	100

Table 2 Performance comparison for ConvNeXt models with different structures

Version	Channels	Modules	MIoU/%	Params/m
ConvNeXt-T	[96 192 384 768]	[3 3 9 3]	74.55	29
ConvNeXt-S	[96 192 384 768]	[3 3 27 3]	74.34	50
ConvNeXt-B	[128 256 512 1024]	[3 3 27 3]	74.06	87
ConvNeXt-L	[192 384 768 1536]	[3 3 27 3]	73.83	198

Table 3 Performance comparison of different neighborhood ranges of k

k	F_1 /%	MIoU/%
3	86.13	77.82
5	86.51	78.31
7	87.11	78.95
9	86.02	77.67

which the ConvNeXt block is stacked within each network layer. Among the versions, ConvNeXt-T yielded the most favorable experimental outcomes on the TTMural dataset. Given that TTMural is a small sample dataset, the network is prone to overfitting when the number of channels is too high and the layers are too deep. And ConvNeXt-T achieved the best objective metrics while having the least number of channels and shallowest depth. Therefore, it is chosen as the backbone network to reduce the possibility of overfitting problems while making full use of the feature data, and to improve the performance of the model in the specific task of segmenting tomb mural cracks.

Keeping other modules and parameters unchanged, we experimentally tested the effects caused by different values of the neighborhood range k in the NA. As shown in Table 3, the performance is optimal when the neighborhood range is 7. When the neighborhood range is small, not enough detailed features can be extracted, resulting in insufficient performance. A neighborhood range exceeding 7 results in extracting excessive irrelevant information, thereby reducing the $MIoU$ value. Therefore, choosing a neighborhood range of 7 avoids over-complexity of the network and ensures the best performance of the model.

Experiments varied the dilation rate of the BA-B branch while maintaining consistency in all other modules and parameters. The results are shown in Table 4, where $MIoU$ increases as the dilation rate increases. This demonstrates that employing multiple branches with varying dilation rates can effectively expand the model's receptive field, thereby enhancing the segmentation

Table 4 Performance comparison of different dilation rates

Dilation rate	F_1 /%	MIoU/%
1	85.67	76.87
3	86.04	77.31
5	86.51	78.01
7	87.11	78.95
9	86.13	77.68
11	85.51	76.78

Table 5 Performance comparison of different segmentation models

Model	P/%	R/%	F_1 /%	MIoU/%	Jaccard/%
U-Net[10]	86.87	80.60	83.62	74.31	52.25
DeepLab V3+[11]	78.82	87.13	82.77	72.84	50.58
DCSAU-Net[20]	86.68	84.05	85.34	76.63	56.67
Swin-Unet[21]	85.12	84.07	84.59	75.72	55.08
UCTransNet[22]	85.65	82.97	84.29	75.96	55.51
DADNet(Ours)	86.81	87.41	87.11	78.95	61.05

performance. When the dilation rate exceeds 7, the model's performance not only fails to improve further but also decreases. This indicates that the dilated convolution cannot be increased indefinitely. If there are too many branches in the dilated convolution, too much irrelevant information is retained, leading to performance degradation. Consequently, setting the model's dilation rate to 7 in practical applications optimizes performance and achieves efficient crack segmentation in tomb murals with minimal computational effort.

Comparative experiments

As shown in Table 5, the model proposed in this paper has a P of 86.81%, a R of 87.41%, a F_1 of 87.11%, a $MIoU$ of 78.95%, and a $Jaccard$ of 61.05% on the TTMural test set. The model attained the highest scores in four out of the five objective metrics evaluated. Meanwhile, the objective metrics achieved by DADNet on the testing set were found to be very close to the objective metrics achieved on the training set 86.92%, 87.83%, 87.37%, 79.26%, 61.44%. This further proves that the model does not suffer from overfitting.

Figure 6 demonstrates that while the P of the DADNet is slightly lower than that of U-Net, it simultaneously achieves high scores in both P and R . This results in an improved F_1 for this method compared to other methods.

Figure 7 illustrates a comparison of the segmentation results obtained from different models. As demonstrated in the first row, our model can segment the crack region

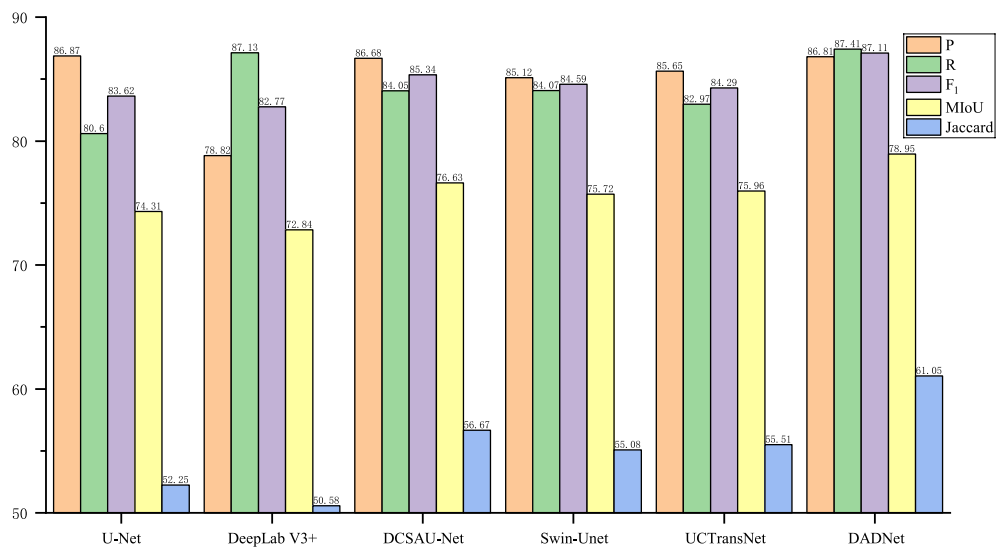


Fig. 6 Performance comparison of different segmentation models

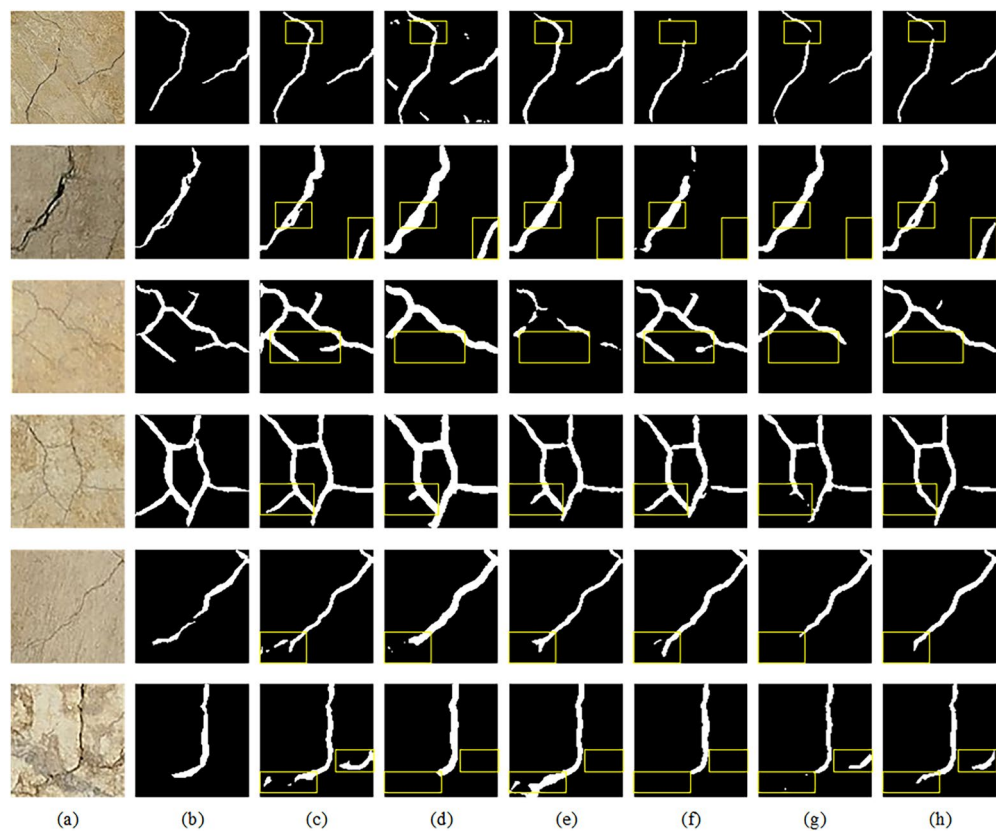


Fig. 7 Visual comparison of segmentation results across models: **a** Input, **b** Mask, **c** DADNet, **d** UCTransNet, **e** Swin-Unet, **f** DCSAU-Net, **g** DeepLab V3+, **h** U-Net

more comprehensively while avoiding segmentation into non-crack regions. The results in the second row show that our model can segment the crack region more accurately. This precision does not result from simply segmenting all areas containing cracks. The results in the third, fourth, and fifth rows demonstrate that our model

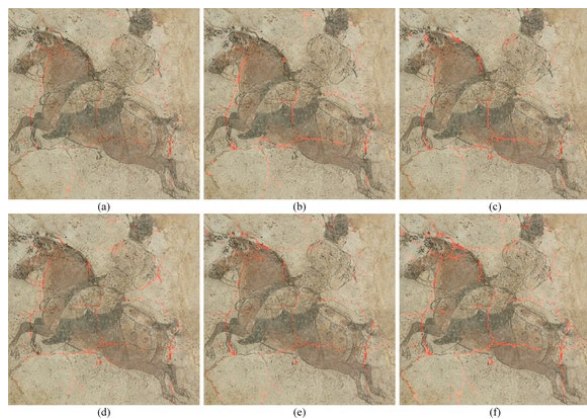


Fig. 8 Segmentation results corresponding to different models: **a** U-Net, **b** DeepLab V3+, **c** DCSAU-Net, **d** Swin-Unet, **e** UCTransNet, **f** DADNet

Table 6 Comparison of model complexity

Model	Params/m	FLOPs/G	FPS
U-Net[10]	24.89	43.19	140.56
DeepLab V3+[11]	54.71	15.94	32.36
DCSAU-Net[20]	2.60	6.92	49.44
Swin-Unet[21]	154.24	119.37	17.00
UCTransNet[22]	65.60	63.20	24.52
DADNet(Ours)	127.30	29.69	27.97

can more accurately segment the smaller craquelures and crack endings regions of the mural. All models accurately segment cracks that are distinguishable from the background. However, for some crack endings and finer craquelures, many models suffer from under-segmentation. The results in the sixth row show that our method can still accurately accomplish the task of crack segmentation even when the cracks are more resemblance to the nearby murals. Overall, these results validate that DADNet excels in the identification and localization of cracks during the segmentation of ancient tomb murals.

To illustrate the actual segmentation outcomes of various models, the segmented areas have been overlaid onto the original image in red. The actual results are shown in Fig. 8. The figure reveals that our method's segmentation results are more accuracy and complete than those of competing methods.

To assess the time and space complexities of various models, we employ three metrics for comparison: Parameters (Params), Floating-point Operations (FLOPs), and Frames Per Second (FPS). Params represent the total count of parameters required for model training, serving as an indicator of the model's space complexity. FLOPs represent the number of floating-point calculations

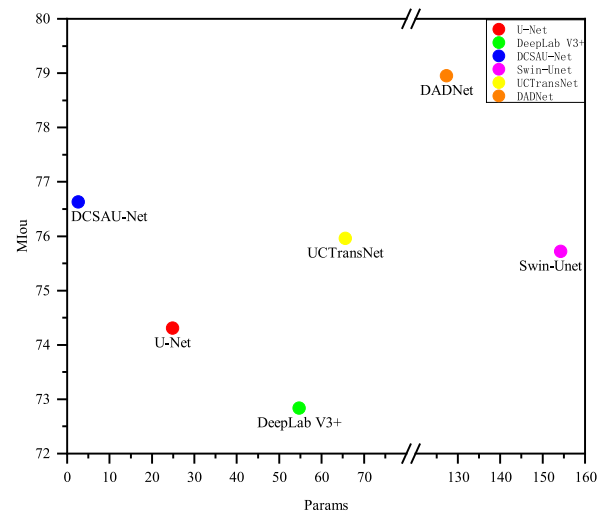


Fig. 9 Model parameters and mean intersection over union

Table 7 Results of ablation study conducted on each module

Model	P/%	R/%	F ₁ /%	MIoU/%	Jaccard/%
U-Net	86.87	80.60	83.62	74.31	52.25
U-Net+ConvNeXt	85.80	81.72	83.71	74.55	52.81
U-Net+BA-B	86.25	82.66	84.42	75.44	54.45
U-Net+NA	86.51	83.89	85.18	76.53	56.35
U-Net+ConvNeXt+BA-B	87.20	85.44	86.31	77.89	59.01
U-Net+ConvNeXt+NA	87.25	85.98	86.61	78.30	59.77
DADNet	86.81	87.41	87.11	78.95	61.05

required during model training. FPS denotes the number of images that the model can process per second. Table 6 presents the Params, FLOPs and FPS metrics of different models using the TTMural dataset. The table reveals that the convolutional U-net model alone has a significant advantage in detection speed, achieving 140.56 frames per second. The DADNet processes images at a rate of 27.97 FPS, equating to 27.97 images per second. In comparison to the Swin-Unet and UCTransNet models, which employ a pure transformer architecture, DADNet is less computationally demanding and provides faster image processing.

Figure 9 illustrates the relationship between model size and performance by plotting model Params against the MIoU.

Ablation study

As shown in Table 7, in the experiments, ConvNeXt is initially integrated with the baseline model to achieve a model with improved performance. This improvement indicates that the ConvNeXt backbone is proficient at

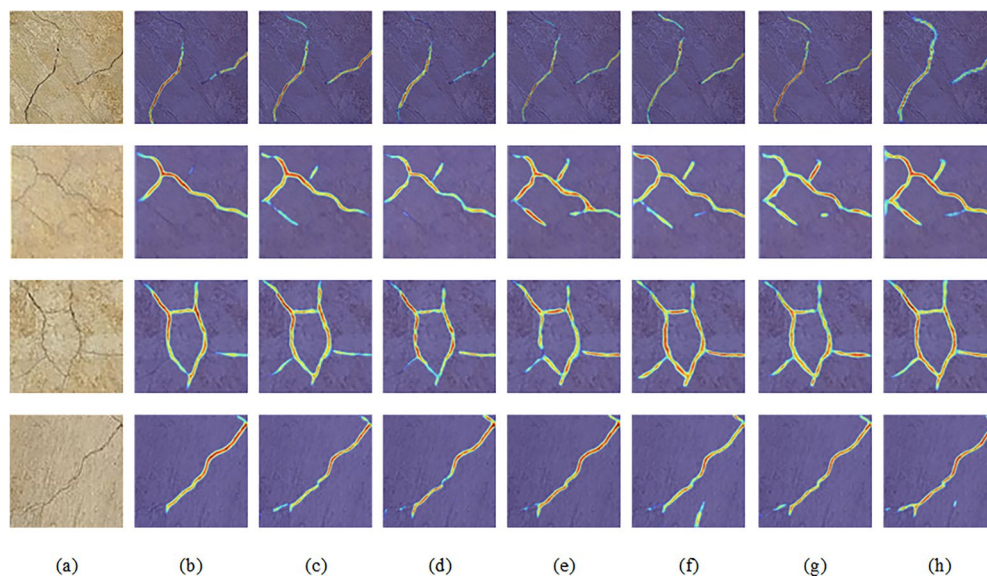


Fig. 10 Comparative results of ablation study: **a** Input, **b** U-Net, **c** U-Net + ConvNeXt, **d** U-Net + BA-B, **e** U-Net + NA, **f** U-Net + ConvNeXt + BA-B, **g** U-Net + ConvNeXt + NA, **h** DADNet

extracting a more extensive range of feature information. The subsequent introduction of the BA-B leads to further performance gains, highlighting the crucial role of the BA-B in extracting advanced semantic and multi-scale information for crack segmentation. Finally, the NA module has been added to the network for optimal performance. The results demonstrate that the local detailed information of cracks captured by the NA module significantly enhances crack segmentation accuracy. These enhancements have a cumulative effect on the model. Each module collaborates with the others to collectively enhance the segmentation capabilities of the DADNet network.

Figure 10 illustrates the impact of each module in the ablation study on the network's performance. The figure shows that the DADNet more effectively focuses on the crack regions. In contrast, U-Net struggles to identify certain smaller craquelures and crack endings regions. Upon adopting ConvNeXt as the backbone network, the system detects an increased number of crack regions. Furthermore, the attention modules adeptly leverage existing feature information to enhance the network's segmentation performance.

Conclusion

In this paper, we present a novel dual-attention detection network model for tomb mural crack segmentation. This model incorporates both the ConvNeXt framework and the DA module in identifying cracks in tomb murals. Moreover, the customized TTMural dataset was constructed by collecting 1,000 mural images and

their corresponding crack annotations. The experimental results demonstrate that the model exhibits specific advantages in segmenting cracks in tomb murals. Utilizing ConvNeXt as the backbone network enables the extraction of more detailed feature information from tomb mural images. The NA makes the model more focused on the local features of the cracks. The BA-B addresses the limitations of NA in extracting global dependencies, thereby enhancing network performance.

Future work will focus on optimizing the structure of the model to reduce its time and space complexities, and to improve the accuracy and robustness of segmentation. To address potential overfitting, we plan to amass a larger and more diverse collection of tomb mural data to expand our dataset. Additionally, we will delve deeper into analyzing crack characteristics and devise new pre-processing techniques to minimize the effect of interference noise on model performance.

Abbreviations

DA	Dual attention
NA	Neighborhood attention
BA	Biaxial attention
BA-B	Biaxial attention block
P	Precision
R	Recall
F_1	F_1 Score
MIoU	Mean Intersection over Union
Jaccard	Jaccard distance
Params	Parameters
FLOPs	Floating-point operations
FPS	Frames per second

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Author contributions

MW designed the idea and led the writing of the paper, RCC and YQZ conducted the experiment, collected the data, and revised the paper. ZYL conducted the analysis and supervised the whole process. All authors read and approved the final manuscript.

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Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

The authors declare that they have no competing interests.

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